

LYON ST EXUPERY AP, France

WMO: 074810

Lat: 45.7264N

Lon: 5.0778E

Elev: 821

StdP: 14.27

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.6	24.9	15.4	12.4	25.6	19.4	15.0	28.3	27.9	49.1	25.1	46.1	6.0	340	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	91.5	68.0	88.0	67.3	84.8	66.5	70.4	85.2	69.1	83.2	67.8	81.1	10.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	98.4	75.0	64.5	93.8	73.6	63.2	89.5	72.7	34.8	85.6	33.6	83.4	32.6	81.4	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
24.1	20.7	17.9	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			17.7	95.9	4.2	3.7	14.6	98.5	12.1	100.6	9.8	102.7	6.7	105.4
			16.9	72.9	4.3	1.5	13.8	73.9	11.3	74.8	8.9	75.6	5.9	76.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	54.9	38.9	40.8	47.7	53.6	60.9	67.9	71.7	71.0	63.4	56.6	46.0	40.0	(6)
(7)		DBStd	13.60	7.67	7.53	6.76	6.53	6.53	6.87	6.24	6.44	6.21	7.08	7.67	7.88	(7)
(8)		HDD50	1290	350	264	126	37	3	0	0	0	24	166	320	(8)	
(9)		HDD65	4322	810	677	538	345	160	46	13	15	102	270	571	775	(9)
(10)		CDD50	3096	5	6	53	144	342	537	672	652	403	228	45	9	(10)
(11)		CDD65	655	0	0	0	3	34	134	220	201	54	9	0	0	(11)
(12)		CDH74	5737	0	0	0	32	264	1204	2025	1802	384	26	0	0	(12)
(13)		CDH80	2028	0	0	0	2	43	406	770	726	81	0	0	0	(13)
(14)	Wind	WSAvg	7.4	7.5	7.8	8.0	8.1	7.6	7.3	6.9	6.6	6.7	7.1	7.3	7.3	(14)
(15)	Precipitation	PrecAvg	36.1	2.4	2.0	2.2	3.1	3.5	2.9	2.8	2.7	3.8	4.3	4.1	2.5	(15)
(16)		PrecMax	48.4	5.3	4.8	6.4	7.4	7.4	6.2	6.9	6.5	11.5	10.4	11.9	5.3	(16)
(17)		PrecMin	27.5	0.7	0.2	0.3	0.5	0.7	1.0	1.1	0.9	0.9	0.8	1.4	0.3	(17)
(18)		PrecStd	5.7	1.2	1.2	1.4	1.7	1.6	1.3	1.5	1.3	2.6	1.9	2.4	1.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.7	61.8	70.3	78.4	84.6	93.4	95.2	97.2	86.4	77.2	67.3	60.1	(19)
MCWB			51.3	49.3	53.8	59.5	64.2	69.0	68.1	68.5	65.5	62.3	56.6	52.7	(20)	
2%		DB	54.9	57.9	66.0	73.0	80.3	88.3	91.2	91.6	82.4	73.6	63.2	56.8	(21)	
		MCWB	48.9	48.7	51.9	56.7	61.7	67.7	68.2	67.9	64.9	61.4	55.0	50.9	(22)	
5%		DB	52.4	54.7	62.4	69.2	76.7	84.7	87.8	87.8	78.6	70.3	60.2	54.2	(23)	
		MCWB	47.3	47.4	51.1	55.1	60.8	66.6	67.6	66.9	63.4	60.3	53.7	49.2	(24)	
10%		DB	49.8	51.9	58.9	65.5	73.2	81.2	84.3	83.8	74.9	67.2	57.2	51.5	(25)	
		MCWB	45.7	45.8	49.8	53.4	59.5	65.3	66.6	66.6	62.4	59.2	52.1	47.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.8	51.4	55.7	60.8	66.5	72.6	72.0	71.7	69.1	64.5	58.3	53.6	(27)
MADB			56.9	57.5	65.9	75.0	79.5	86.2	87.0	87.5	80.6	72.6	64.3	58.1	(28)	
2%		WB	49.5	49.6	53.6	58.1	64.0	69.9	70.2	70.0	66.6	62.9	55.9	51.6	(29)	
		MADB	53.9	56.1	62.5	69.9	76.5	83.8	85.4	84.9	77.5	70.6	61.5	56.3	(30)	
5%		WB	47.9	48.0	52.0	56.1	62.3	68.0	68.9	68.7	65.1	61.4	54.2	49.8	(31)	
		MADB	51.8	53.9	60.4	66.7	73.1	81.3	83.4	82.7	75.2	68.1	59.4	53.6	(32)	
10%		WB	45.9	46.3	50.4	54.3	60.5	66.4	67.5	67.4	63.5	59.8	52.5	47.6	(33)	
		MADB	49.4	51.6	58.1	63.7	70.4	78.5	81.0	80.4	72.6	66.0	57.0	51.0	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.7	12.4	15.9	16.7	16.8	18.0	18.7	17.0	14.2	11.1	9.5	(35)	
MCDBR			12.8	18.0	21.3	22.4	22.0	22.6	23.1	24.2	22.2	18.0	15.0	12.8	(36)	
MCWBR			8.5	10.8	10.7	10.4	9.3	8.6	7.9	7.9	8.7	8.4	9.1	8.5	(37)	
5% WB		MCDBR	12.1	16.1	18.7	20.2	19.3	21.1	20.7	21.1	18.7	15.8	13.7	11.7	(38)	
		MCWBR	8.4	10.1	10.1	10.0	8.9	8.5	7.8	7.8	8.2	8.0	8.8	8.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.309	0.329	0.359	0.395	0.401	0.402	0.399	0.392	0.376	0.361	0.329	0.306	(40)	
(41)		taud	2.476	2.414	2.353	2.259	2.290	2.322	2.328	2.355	2.384	2.425	2.472	2.482	(41)	
(42)		Ebn at Noon	254	269	275	274	275	275	274	272	266	253	243	242	(42)	
(43)		Edh at Noon	22	28	34	41	41	40	39	36	32	27	22	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	396	665	1078	1434	1674	1994	1954	1688	1289	805	446	332	(44)	
(45)		RadStd	47	94	132	184	159	159	166	126	100	80	50	56	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page